

Where Does Agency Live?

Nondeterminism as a computational resource, and agency across the computational and biological axes

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Abstract

The relationship between agency and determinism is an old question; this note brings a new instrument to it. We plot the phenomena on two axes — computation and biology — and use each to calibrate the other. The analytical hinge is a single reframing: in a model of computation where interaction is primitive, the nondeterminism of an interaction is not missing information but a *resource*, namely a choice. Made precise, that choice is a choice function for a definite family, so the mathematics of the axiom of choice acquires a computational meaning, and the strength of the choice principle one spends becomes a coordinate in a hierarchy of computational power — a stalk in a *fibration of hypercomputation* over the category of models of computation and their bisimulation-preserving morphisms. The mobile process calculi sit in this category as the objects built expressly for agency: they take all computation to arise from interaction among agents, and they are *fundamentally non-confluent*, which is exactly what is needed to speak of races — two sodium atoms for one chlorine, two sperm for one egg, two predators for one prey, two buyers for the last ticket. RSpace, an abstraction that lets these calculi run in production, is where the choice implicit in the nondeterminism becomes visible. We then turn the instrument on agency itself: once choice is localised, the question of where to draw an agent’s boundary gets a fresh answer, and Eric Smith’s reading of the biosphere — the agents that eat outside energy as the true interface, the agents that eat each other as a determinate interior — becomes a computational factorisation. The interior is determinate not because it cannot branch but because its branching does not matter: it is confluent, in the way the λ -calculus is. We call such a subsystem a macro-component (an ecosystem is the candidate), and observe that macro-components meet at cuts where nondeterminism is real. The fibration, finally, is a map and not the territory: the actual universe of computational phenomena may resemble the physical one, with scales and capacities wildly juxtaposed — femtosecond molecular dynamics beside minute-scale gene expression in a single cell, halting oracles beside λ -terms at a single cut. That juxtaposition is the flat ontology, and biology is our reason to take it seriously.

1 An old question, two axes

Is agency compatible with a determined world, and if there is room for agency, where exactly is the room? The question is old and the usual moves are well-worn. We want to bring a different instrument: the theory of computation, sharpened by a particular family of models in which

interaction is the primitive and nondeterminism is not noise but structure — and, in parallel, the biology of living matter, where agency is not a philosopher’s posit but the daily business of cells, organisms, and ecosystems. We treat these as two axes on which to plot the same phenomena, and the discipline we hold ourselves to is bidirectional: every claim should read in both directions. Computation should inform our picture of biological agency, and biology should inform our picture of what computation is. The flat ontology we arrive at is motivated precisely by that second direction — the living world shows us, concretely, the juxtaposition of scales and capacities that the computational picture, left to its own neat hierarchies, would not have predicted.

The fresh move is small to state and large in consequence. In a model of computation built on interaction, when two agents meet and could combine in more than one way, the selection of *which* way is a genuine degree of freedom. The standard reflex is to treat that degree of freedom as something to be averaged away or refined out — nondeterminism as ignorance. We treat it instead as a *resource*: a choice, in the precise mathematical sense, whose supply has a cost and whose strength can be measured. Once nondeterminism is a resource, two things follow that organise the whole essay. First, the discourse of the axiom of choice — usually filed under foundations of mathematics — becomes computationally load-bearing: different ways of supplying choice import different amounts of computational power. Second, the question of agency relocates. Agency lives where the choice is open, and the boundary of an agent is best drawn around the cuts where its choices are genuinely live. Not every place two agents meet is such a place; much of what looks like interaction is determinate traffic with no choice to be had. Finding where the choice actually is turns out to be the same as finding where agency actually is.

2 A landscape of computational models

To make “a coordinate in a hierarchy of computational power” precise we need a space of models, not a single model. Fix the following landscape, briskly.

Objects: models of computation as graph-structured lambda theories. A wide range of computational formalisms — the λ -calculus, term-rewriting systems, Petri nets, the process calculi — can be presented uniformly as *graph-structured lambda theories* (GSLTs): a syntax of terms with a binding discipline, a notion of context, and a reduction or interaction relation, all organised so that the term structure is a graph rather than merely a tree [5]. The details of the presentation are not needed here. What we need is that each such model carries a canonical behavioural equivalence — *bisimilarity* — generated, in the GSLT discipline, by labelling transitions not with primitive actions but with the minimal contexts that enable them, after Leifer and Milner [8, 9]. Two computations are bisimilar when no context can tell them apart; the observer-specified logic functor (OSLF) of [5] generates, from the same context labels, the modal logic whose formulae they satisfy, so behavioural equality and logical indistinguishability coincide.

Morphisms: maps that preserve bisimulation. Let GSLT be the category whose objects are these models and whose morphisms are the maps that preserve bisimilarity — translations of one model into another that respect when two computations are behaviourally the same. This is the right notion of morphism for our purpose, because the thing we will fiber over each model is its *observable behaviour*, and behaviour is exactly what bisimilarity tracks. A

bisimulation-preserving morphism carries a model’s behavioural fabric, and hence its hierarchy of behaviours, into another’s.

The fibration of hypercomputation. Over the single Turing-complete model, Turing already gave us a tower: relativise computation to an oracle, take the halting problem of the relativised machine as a new oracle, iterate, and index the iteration by the ordinals [10] — the arithmetical hierarchy and the Turing degrees, the classical apparatus of hypercomputation [11, 12]. The instrument we want is the *generalisation of this tower to a tower per model*. Over each object G of GSLT stands not a single computational power but a whole hierarchy — G relativised to its own oracles, the hypercomputation tower *specific to that model of computation*. These towers assemble into a fibration

$$p: \text{Hyp} \longrightarrow \text{GSLT},$$

whose fiber (stalk) over G is the hypercomputation hierarchy native to G , and whose vertical direction is added computational power. Reindexing along a bisimulation-preserving morphism $G' \rightarrow G$ — the cartesian direction — carries a high-power computation to its shadow in a weaker frame, which is exactly the act of describing a strong system in a weaker vocabulary. We will lean on this picture twice: first to give “choice as a resource” a vertical coordinate, and later, when we ask whether the fibration is the territory or only a chart of it.

3 Interaction, and why non-confluence is the point

Among the objects of GSLT, one class is built for our question. The mobile process calculi (MPCs) — Milner’s π -calculus and, in the form we use, the reflective higher-order (rho) calculus [1] — take a definite philosophical stance: *all computation arises from interaction among agents* [7]. There is no lone machine grinding a tape; there are processes that meet, exchange, and continue, and computation is what happens at their meetings. We write the meeting as a parallel composition, the *cut*:

$$P \mid Q,$$

and the single base reaction is communication across it,

$$\text{for}(y \leftarrow x) P \mid z!(Q) \longrightarrow P\{\text{@}Q/y\} \quad (x \equiv_N z),$$

an input on a channel meeting a matching output. Agents flank every cut; the cut is where they act on one another.

The feature that makes the MPCs the right tool — and that distinguishes them sharply from the λ -calculus — is that they are *fundamentally non-confluent*. When several outputs and several inputs collect on the same channel, which output meets which input is not determined by the terms; it is a real choice, and different resolutions lead to genuinely different, non-reconvergent futures. This is usually presented as a complication. We present it as the entire point, because non-confluence is the computational signature of a *race*, and races are how agency shows up in the world. Consider, across four scales:

- two sodium atoms competing to donate their electron to one chlorine;
- two sperm racing to fertilise one egg;
- two predators converging on a single prey;

- two buyers reaching for the last concert ticket.

Each is, structurally, one consume-slot (the chlorine, the egg, the prey, the ticket) contended by several produce-candidates, resolved by a pairing that the situation does not fix, with the unmatched candidates left over. Each outcome *matters*: this sodium and not that one is now bound; this sperm and not that one founds the lineage. A confluent model cannot speak of these, not because it lacks the vocabulary but because confluence is precisely the claim that the choice does not matter — that all resolutions reconverge. The λ -calculus, the great confluent model, is the stakes-free corner of the landscape: its many reduction orders are real choices whose outcomes are identified by Church–Rosser, so the choice is free and invisible. Reality is mostly not like that. To talk about agency one needs a model in which the choice can matter, and the MPCs are the objects of GSLT in which it does. This is the first crossing of our two axes: biology (the race for the egg) tells us which computational models are adequate to agency, and the answer is the non-confluent ones.

The rho calculus, specifically. We use the rho calculus because its names are quoted processes — $@P$ turns a process into a name, $*x$ turns a name back into a process, and $*@P \equiv P$ — so there are no primitive names and no name-restriction operator; reflection furnishes recursion with no primitive replication [1]. Nothing below turns on the set-theoretic models of [2, 4]; we draw only on the shape of the calculus. The reader may keep the modern surface syntax in view: $@P$ for quote, $*x$ for dereference, $\text{for}(y \leftarrow x) P$ for input, $x!(Q)$ for output, $|$ for the cut, $\equiv$ for structural congruence, $\sim$ for context bisimilarity (the behavioural, congruence-by-construction equivalence) and $\sim_r$ for resource bisimilarity (the finer equivalence that counts parallel copies).

4 RSpace: the bridge, and where the choice is

The MPCs are admirable for reasoning and awkward to run. RSpace is the abstraction that closes the gap: it presents the parallel fragment of the rho calculus as a store — a hashtable from channels to polarity-homogeneous bags, either of data (produces) or of continuations (consumes) — and interprets the cut as keyed multiset union, so the denotation $\llbracket - \rrbracket$ is a monoid homomorphism and the structural laws of $|$ become literal equalities of tables [2]. This is not a toy: it is how the calculus is compiled into a high-performance, persistent, content-addressed store in production. We invoke it here for one reason: it makes the choice of §3 *visible and operational*.

When a channel carries both a bag of consumes and a bag of produces, communication chooses an underlying list of each and *zips* them — matching pairs react, unmatched payloads remain as surplus at the channel. The denotation stays a function; the zip is the separate dynamics; and the choice of which list orderings to zip is the calculus’s nondeterminism, now pinned to a key. The four races of §3 are this forgiving zip with surplus, drawn from life: several candidates, one slot, a pairing the data does not fix, leftovers remaining. RSpace’s contribution to the present inquiry is to show that the nondeterminism is not a haze over the system but a specific, locatable act of selection — and an act of selection is a choice, in the sense the next section makes exact.

Remark 1 (A motivating example, not a mechanism). We are not proposing that sodium, sperm, or ticket-buyers *run* RSpace, or that its store is a physical mechanism. The claim is structural: the abstract shape — contended slot, nondeterministic pairing, surplus — is shared, and RSpace

is where that shape, and the choice inside it, is displayed cleanly enough to analyse. The value of the MPC and RSpace vocabulary is that it *names the categories*; whether any given physical system instantiates them is a separate, empirical question we do not prejudge.

5 Choice as a computational resource

Now make the choice exact and connect it to the stalks of §2. Index by the channels carrying mixed polarity (or, for a run, by the reaction events). At each index the fibre is the set of admissible pairings — inhabited precisely when a redex is present. A completed schedule is a section of this family: an element of the product of the fibres. The assertion that a section exists is an instance of the axiom of choice, and the scheduler that realises it is, literally, the choice function for the pairing family. Nondeterminism resolved *is* choice exercised. This is the practical reframing of AC: not a remote axiom about arbitrary sets but the principle that supplies a scheduler, and its various restrictions and strengthenings are different schedulers with different powers.

The strength required is graded, and the grading is the one the source calculus’s risk ledger already keeps [2].

- *Finite contention, finite run.* The index set is finite and the product is inhabited with no choice axiom at all — constructively. More: a bag is a parallel composition of finitely many prefixes, so it arrives *presented* as a list-up-to-permutation. The term hands you the enumeration; choosing a linearisation is free. In the finite world choice is not non-constructive; the term is its own choice structure.
- *One ω .* Reflection (or non-termination) installs an ω -fold contention, or an ω -sequence of events whose pairings depend on the residuals of the last. The static case needs countable choice; the dependent case needs Dependent Choice. This single ω is the one the ledger spends; here it buys exactly the step from no-choice to countable/dependent choice.
- *Class size.* Full AC, and only here.

The decisive connection is vertical. A scheduler that supplies only finite, demonic choice keeps the computation at the base of its hypercomputation stalk: it imports no power the model did not already have. A scheduler that can survey an infinite contention and pick a globally coherent matching — an angelic or oracle-endowed scheduler — imports computational power, and that power is altitude in the fiber over the model. The cleanest way to say it: *choice is the vertical coordinate*. Where you sit in the hypercomputation stalk of §2 is set by how strong a choice principle your scheduler is allowed to spend. The Weihrauch lattice [13], whose objects are choice principles ordered by uniform reducibility and composed by “do one, then the other,” is the natural home for this gradation — and, tellingly for what follows, it is a *lattice*, not a linear chain. This is the second crossing of our axes, now internal to computation: the foundational mathematics of choice turns out to measure the same thing as the theory of relative computability, because both are measuring the resource a race consumes when it is resolved.

6 Choice and the drawing of boundaries

We can now turn the instrument on agency. If nondeterminism is localised at cuts and is a resource, then the natural place to look for an agent’s boundary is where its choices are live.

Make two notions precise.

Definition 1 (Cut and aperture). A *cut* is a parallel composition co-locating a consume and a produce — the locus where two agents interact. A cut is an *aperture* when the resolution of its pairing nondeterminism is *observable*: distinct admissible pairings lead to residuals in distinct behavioural classes (distinct $\sim_r$ -classes at the resource layer, or distinct $\sim$ -classes after forgetting multiplicity). A cut that is not an aperture carries no observable choice: either there is a unique match, or the several matches reconverge.

Principle 1 (Not every cut is an aperture). Agents flank every cut, but only apertures leave the interaction’s nondeterminism open. And only at an aperture can an agent of higher computational capacity influence the behaviour of a lower one — by biasing the live choice. Where a cut is not an aperture there is no choice to bias, and the stronger agent’s surplus power finds no purchase. Agency, in the sense of room-to-do-otherwise, lives at the apertures.

This is already a fresh answer to the old question. The apertures are exactly the races; the determinate traffic between them is not where agency is decided, however much computation flows through it. And it locates interference precisely: a high-capacity agent reaches into a low-capacity one not everywhere they touch but only at the cuts where the low-capacity one has a live, observable choice that the high-capacity one can see how to resolve. The asymmetry of capacity — one agent several halting-oracles above another — becomes operative only through an aperture.

Two corrections to a tempting oversimplification, both of which we will need. First, the aperture is a property of the *cut*, not of either agent; nondeterminism is not preserved by $|$, so a determinate whole can have wildly underdetermined internal cuts, and conversely. “An agent’s nondeterminism budget is its interface” is therefore not quite right: the budget belongs to the cut, and which cut one calls the boundary is a modelling decision. Second, because of this, a seam where choice is genuinely open may sit far from the boundary an observer would draw, and its effects may be felt at cuts the observer counts as internal. We turn to a setting where this plays out concretely.

7 The biological axis: Smith’s biosphere

Smith and Morowitz read terrestrial life as a planetary process and ask after its interface to the rest of the universe [6]. On their account the agents that matter at that interface are the ones that consume energy from *outside* the biosphere — sunlight, the chemical disequilibria of hydrothermal vents — while a vast interior of agents consume *one another*, trading energy already captured. The energy-eaters are the actual boundary between (terrestrial) life and the cosmos; at that level the agentic boundary makes the entire biosphere a single agent. Yet humans, themselves alive, recognise a far finer delineation — bees distinct from birds, this bee distinct from that.

The MPC vocabulary names these categories cleanly. Schematically,

Biosphere $\equiv$ EnergyEaters $|$ AgentEaters, Universe $\equiv$ Biosphere $|$ RestOfUniverse.

The seam to the cosmos is the cut Biosphere $|$ RestOfUniverse, and EnergyEaters is the sub-term whose business is that cut: the place where outside energy — untapped to any particular

molecule, the sun not addressing its photons to a chosen chlorophyll — enters as an underdetermined produce, a genuine aperture. *AgentEaters*, by contrast, may be, taken as a whole, determinate. This is so *even though* its internal interactions are exquisitely constraint-sensitive: the bee consumes the pollen, not the worm the bird eats; enzymes are spectacularly specific, one substrate per active site. The interior obeys a thicket of conformance constraints and yet runs no races — the constraints do not select among matches, they ensure there is only one match to select.

Remark 2 (Two senses of “agent”). The bees are real agents to each other and to us — distinct nodes, distinct types, distinct roles. But they need not be *sites of choice*. Their distinctness is type-distinctness, not aperture-distinctness. The only underdetermined thing in a bee’s life may be which upstream race gates the arrival of its pollen; given that arrival, the rest is forced. So discrimination among agents and the location of nondeterminism come apart: an agent in the *discrimination* sense (an individuated role) is not, in general, an agent in the *choice* sense (a cut with a live race). The finer, bee-level description sees *more agents* but — if the interior is determinate — the *same apertures*, all at the macro-boundary.

This is exactly the seam-versus-boundary phenomenon of §6, now biological. The open choice sits at the energy cut; its effects are felt at the predation cuts, deep inside; and at that depth the agents may experience an externally sourced bias as their own internal, determinate dynamics — unable, from inside their own capacity, to distinguish “my choice resolved thus” from “the external write reached down to here.” The seam genuinely violates the boundaries an observer would draw, entering at one level and metabolised, at another, into what looks locally like determinism. Whether that “looks like” is exact — whether the interior really transmits the boundary decision without adding choices of its own — is a computational question, and we now answer it.

8 Back to computation: confluent macro-components

We want a name for the object Smith’s interior instantiates: a subsystem all of whose internal cuts are sealed, with apertures only at its boundary. The first guess is that the interior is sealed by *rigidity* — a unique match at every internal cut, so the zip is a function and there is no race. Syntactically that is *linearity*: each internal channel used once per polarity, each consume discharged by exactly one produce, the boundary the residue of the contended channels. But linearity is too strong. What we actually require is only that the interior carry no *observable* choice, and the exact statement of that is confluence.

Definition 2 (Macro-component). A subsystem is *rigid* if its internal reaction is confluent up to resource bisimilarity: every internal race reconverges, all maximal internal reaction sequences reaching $\sim_r$ -equal normal forms. A *macro-component* is a maximal rigid subsystem; its internal cuts are non-apertures and its boundary cuts — where internal confluence fails — are its apertures. The locus of confluence-failure is the *rigidity frontier*.

Confluence permits the branching linearity forbids. A bee may have several internally admissible pairings; if all flow to $\sim_r$ -equal normal forms it is no site of observable choice, which is all the factorisation needs. Metabolism is full of internal branching — redundant pathways, order-independent cascades — that converges on the same steady state: confluent, not linear. The slogan is therefore *the interior is λ -like, not linear-like*. The λ -calculus is the witness: operationally

awash in branching, one reduction order against another, and famously determinate because Church–Rosser makes the branching fictional — one normal form, reached however you like. A *closed* λ -term is a sealed macro-component at every scale, nothing to interfere with because no channel is contended; an *open* term is the agent with apertures, its free variables the boundary channels where a context can substitute and change the result. In the MPC vocabulary the free variable is the unmatched consume, the dangling polarity — the port where the world reaches in.

We propose, tentatively, that *ecosystems are the candidate macro-components*. An ecosystem’s internal web of who-eats-whom settles, up to the bookkeeping that counts copies, to characteristic steady states; its branching — which individual predator takes which individual prey — typically does not change the behavioural class of the whole, and is thus confluent up to resource bisimilarity. Its live races — competition at the edge for the external influx, or genuinely contended predation — are its apertures, and they sit at its boundary with neighbouring macro-components and with the energy source. Macro-components, on this reading, are the determinate interiors; the cuts *between* them are where the nondeterminism of §3 is real. Three frontiers then coincide, which is the structural dividend:

Proposition 1 (One frontier, three readings). *For a macro-component, the rigidity frontier (where internal confluence up to $\sim_r$ fails) coincides with the detectable-interference frontier (where a biased pairing yields a different $\sim_r$ -class, so an external write leaves a mark) and with the confluence-obligation frontier (where the eager-confluence obligation of the path semantics [3] is non-trivial). The place where the interior can no longer hide its schedule is the place where interference becomes observable, which is the place where the formal obligation bites.*

A consequence worth isolating: a confluent interior carries no choice resource — no observable nondeterminism, hence no Weihrauch content, hence (§5) no altitude in the hypercomputation stalk — however baroque its branching. So the resource degree of a macro-component is carried entirely by its boundary, $\deg(\text{macro-component}) = \deg(\text{boundary})$. Capacity, like choice, lives at the apertures.

9 The flat ontology

The fibration of §2 is a map, and we should not mistake it for the territory. It is a way to *understand* computational phenomena — to sort agents by capacity and read interference as reaching down a tower. But the actual universe of computational phenomena may be far less tidy, and biology is the reason to suspect so. In a single living cell, fast molecular dynamics — vibrational and electronic transitions down to the femtosecond — coexist with transcription and translation that unfold over seconds to minutes: many orders of magnitude apart, intimately coupled, in one small volume. The cell does not run its fast and slow processes in separate strata that communicate through a clean interface; it juxtaposes them, and the juxtaposition is the point. The physical universe at large is the same — scales and scopes wildly interleaved rather than cleanly layered.

We propose to take computation the same way. In the *flat ontology*, the universe of computational phenomena is the total space of the fibration *without privileging the projection* — a jumble of agents of wildly different computational capacity, several halting-oracles up beside fully deterministic λ -like agents, interacting directly at cuts. The tower is not the structure of the world; it is a cleavage, a chosen gauge for describing a frame-independent jumble, and the

“stratification” is an artefact of which morphism we chose to project along. Read this in both directions, as promised. Computation, taught by biology, stops expecting a neat hierarchy and starts expecting juxtaposition. Biology, taught by computation, gains a vocabulary for what the juxtaposition costs and where it can bite: a high-capacity agent can influence a low-capacity one, but — Principle 1 — only at an aperture, only where the lower agent has a live choice the higher one can resolve.

What keeps such a flat universe coherent, rather than collapsing to the top capacity? Two disciplines, both already in the MPC vocabulary. *No implicit interaction*: the store never reacts with itself; a reaction is triggered only by a cut that actually co-locates opposite polarities [3]. Drop it and a single oracle on a universal channel homogenises the field upward; keep it and interference must be *cut-triggered*. *Locality and cost*: a strong agent can write only where it shares a channel — and, in the path reading, only into the subspace below a prefix it holds — and a cost discipline makes its reach finite per unit resource. Locality plus cost is what lets agents of disparate capacity share a world without the strongest simply absorbing the rest. These are not bookkeeping; they are the consistency conditions for a flat ontology of unequal agents, and they are exactly the conditions under which the biosphere’s determinate interior can persist beneath an energy-eating, oracle-flavoured boundary without dissolving into it.

10 The hinge: is determinism stable under interference?

One question decides whether the picture is static or dynamic, and it is the question a formalisation should resolve first. Everything above treats “determinate interior, nondeterministic boundary” as a property a system *has*. Whether it *keeps* the property when a boundary aperture fires and the decision propagates inward is not obvious — because confluence up to resource bisimilarity, unlike linearity, is not compositional, and because a deep interaction in the path semantics manufactures a fresh, deeper interaction whose confluence is a new obligation [3].

Question 1 (Stability of rigidity). *Is confluence up to resource bisimilarity preserved as a boundary decision cascades into the interior? Equivalently: when a macro-component’s aperture fires, does its interior stay rigid?*

The two answers are both meaningful, and the choice between them is what biology and computation are jointly asking. If rigidity is stable, the factorisation is robust: the boundary race fires, the decision descends as a *forced* cascade through an interior that has no opinion of its own, and “determinate interior / nondeterministic boundary” is a genuine structure theorem holding along the dynamics — the predation web metabolising the day’s sunlight into ordinary determinate life. If rigidity is *not* stable, then boundary interference can crack open a previously determinate interior, turning a confluent agent into a branching one by descending into it. That is not merely a defect; it is plausibly the formal content of *perturbation* — a high-capacity agent not only steering a low-capacity one within its existing freedom but *manufacturing* new freedom in it and then steering that. In the ontological register: the difference between a universe whose agents have fixed capacities and one in which capacity itself is something agents can do to one another. We do not know which holds, and the same confluence lemma settles it on both axes at once.

11 Open problems

We collect the load-bearing uncertainties, each a structural claim one can try to settle. *(i)* Make precise the family-of-pairings whose section is the schedule and prove the AC-grading of §5, identifying which schedulers are confluent up to $\sim_r$ (constructive, no imported oracle) and which import altitude in the stalk. *(ii)* Exhibit the fibration $p: \text{Hyp} \rightarrow \text{GSLT}$ rigorously and test whether angelic and demonic resolution are the Lawvere adjoints $\exists \dashv (-)^* \dashv \forall$ to reindexing — the decisive sub-question being whether the GSLT fibres carry the limits and colimits both adjoints require. *(iii)* Show “maximal rigid subsystem” (Definition 2) is well defined and that the three frontiers of Proposition 1 coincide, and establish that a confluent interior is resource-degree-transparent. *(iv)* Resolve Question 1 — the stability of rigidity under descent — proving the structure theorem if it holds, or characterising perturbation as a property of the sub-system if it fails. *(v)* Test the proposed reading of ecosystems as macro-components against a concrete ecological model, and locate its apertures.

12 Conclusion

We asked where agency lives, and answered with a computational coordinate: at the apertures, the cuts where nondeterminism is open and observable, which are the races. The route there was a reframing — nondeterminism as a resource, a choice in the exact sense, graded by the axiom of choice and climbing, with its strength, the stalks of a hypercomputation fibration over the category of computational models and their bisimulation-preserving morphisms. The mobile process calculi earned their central place by being fundamentally non-confluent, hence able to speak of races that matter — two atoms, two sperm, two predators, two buyers — and RSpace earned its place by making the choice inside those races visible and runnable. Smith’s biosphere showed the categories at work in life: an energy-eating boundary that is a genuine aperture, an agent-eating interior that is determinate not by rigidity but by confluence, in the manner of the λ -calculus. We named such an interior a macro-component, proposed the ecosystem as its biological instance, and found capacity and choice alike concentrated at its boundary.

The deepest claim is the flat ontology, and it is where the two axes most need each other. The fibration is a chart; the world it charts may be, like the physical and the living world, a juxtaposition of scales and capacities rather than a tower — halting oracles beside λ -terms, femtosecond chemistry beside minute-scale gene expression, all at the same cuts. Computation, taught by biology, learns to expect this; biology, taught by computation, learns what it costs and where it can bite. And the single open hinge — whether a strong agent, reaching through an aperture, can crack open the determinism of a weaker one — is one confluence lemma, which would tell us, in one stroke and on both axes, whether agency is a fixed allotment or something agents continually make and unmake in each other.

Acknowledgements. This essay draws on the RSpace denotation of the rho calculus [2, 3] and the GSLT/OSLF programme [5], and on Smith and Morowitz’s reading of the biosphere [6]; its conceptual claims are deliberately routed back to structural obligations where they can be settled.

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